

Ahead of Their Time

War brought innovation: sometimes brilliant, sometimes clutching at straws, always desperate to win an advantage. A recurring theme on both sides of the conflict (and of the Atlantic) was the “flying wing”—designers’ attempts to incorporate the “dead” weight of fuselage and tail into a thick wing section so that the whole aircraft contributed to its lift and aerodynamic efficiency. The ultimate example, the B-2 Spirit “Stealth Bomber,” would not be revealed until 40 years later.



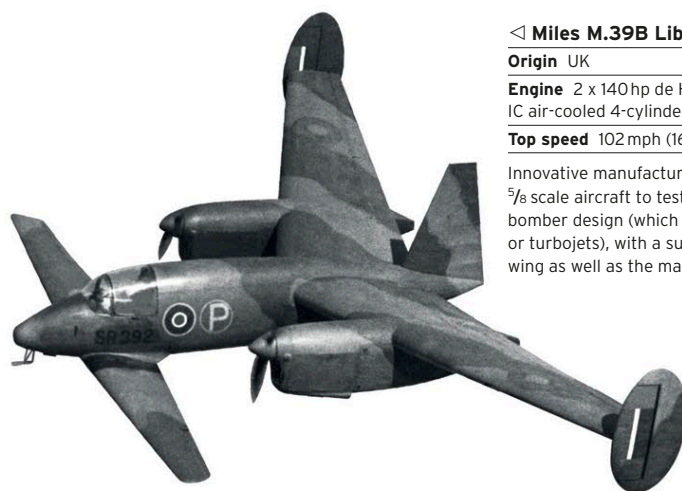
△ Northrop XP-56 Black Bullet 1943

Origin US

Engine 2,000 hp Pratt & Whitney R-2800-29 air-cooled 18-cylinder radial

Top speed 465 mph (749 km/h)

This revolutionary fighter-interceptor was conceived in 1939, as a way to minimize drag with minimal fuselage, diminutive tail, magnesium alloy construction, and an H-24 layout engine. It was unstable.



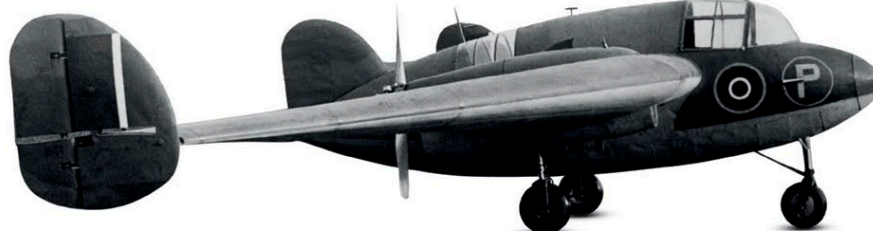
◁ Miles M.39B Libellula 1943

Origin UK

Engine 2 x 140 hp de Havilland Gipsy Major IC air-cooled 4-cylinder inverted inline

Top speed 102 mph (164 km/h)

Innovative manufacturer Miles built this 5/8 scale aircraft to test its revolutionary bomber design (which would have had Merlins or turbojets), with a supplementary low front wing as well as the main one; it flew well.



▽ Handley Page HP75 Manx 1943

Origin UK

Engine 2 x 140 hp de Havilland Gipsy Major air-cooled 4-cylinder inverted inline

Top speed 150 mph (241 km/h)

First conceived in the 1930s, the Manx had part-swept wings, two “pusher” engines, and wing-tip rudders. It was poorly constructed by Dart Aircraft and unable to fly until 1943; tests ended in 1946.



△ Westland Welkin Mk1 1944

Origin UK

Engine 2 x 1,233 Rolls-Royce Merlin 76/77 supercharged liquid-cooled V12

Top speed 385 mph (620 km/h)

Welkin had a pressurized cabin, heated screen, oxygen, and huge wings, to fly interceptor missions at 45,000 ft (13,716 m). It was made in small numbers because high altitude bombing was no longer a great threat.



△ Northrop N-9M Flying Wing 1942

Origin US

Engine 2 x 300 hp Franklin XO-540-7 supercharged air-cooled flat-8

Top speed 258 mph (415 km/h)

Jack Northrop built four one-third scale models of his proposed flying wing heavy bombers in order to test their flying characteristics and familiarize pilots with the design. The project finally came to fruition in form of the “Stealth Bomber” in 1989.

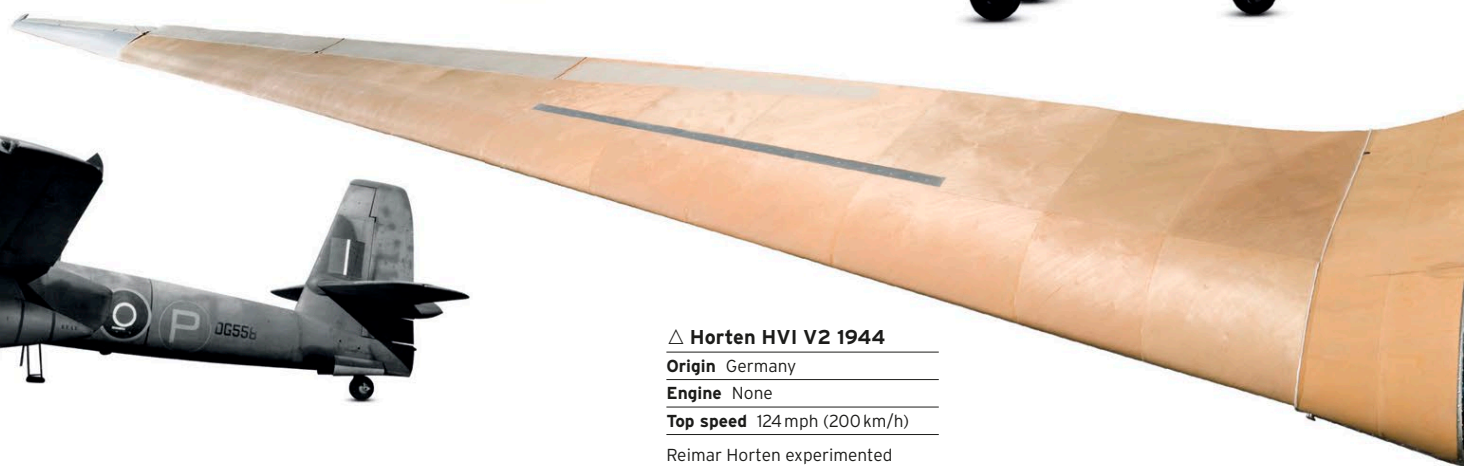
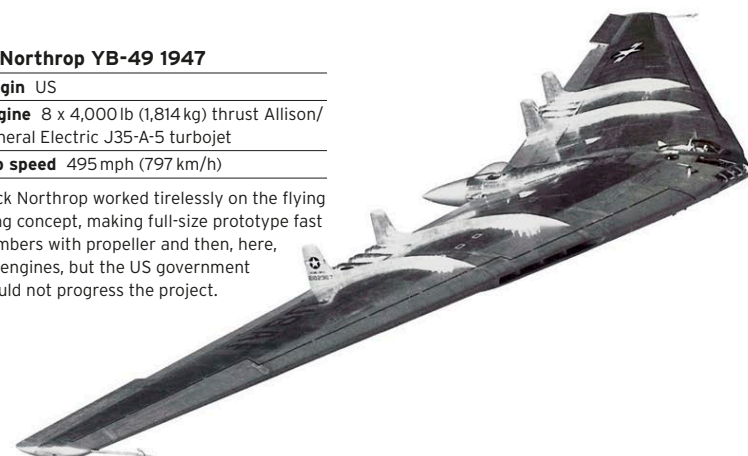
▷ Northrop YB-49 1947

Origin US

Engine 8 x 4,000 lb (1,814 kg) thrust Allison/General Electric J35-A-5 turbojet

Top speed 495 mph (797 km/h)

Jack Northrop worked tirelessly on the flying wing concept, making full-size prototype fast bombers with propeller and then, here, jet engines, but the US government would not progress the project.



△ Horten HVI V2 1944

Origin Germany

Engine None

Top speed 124 mph (200 km/h)

Reimar Horten experimented with swept-wing aircraft that had no fuselage or tail. This glider's pilot had to kneel semiprone and needed oxygen, pressurization, and heated gloves to fly at high altitude.